LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Engineering Metrology, need for inspection, precision and accuracy
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Engineering Metrology, need for inspection, precision and accuracy.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of standard measuring instruments; Safety: Handle glass/plastic scales gently; Supervision: Instructor monitors handling of fragile gauges.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of engineering metrology?
(b) Why is inspection necessary in aircraft maintenance?
(c) Can a measurement be accurate but not precise? Explain.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Engineering Metrology • Scope and applications in aviation • Role of measurement in quality control • Basic units of measurement • Importance of standardization	L & Ds	PPT, IB	15 min
(b)	• Need for inspection in engineering • Difference between precision and accuracy • Sources of measurement errors • Significance of tolerance limits • Impact of errors on aircraft safety	L & Ds	PPT, PDF	20 min
(c)	• Classification of measurement errors • Methods to improve measurement reliability • Role of calibration in metrology • Common instruments used for inspection • Summary of key concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Define precision and provide an example relevant to aircraft parts.
(b) Explain how inspection prevents catastrophic failures in engineering systems.
(c) Differentiate between accuracy and precision using a target analogy.

12. Home Assignment and Follow Up	Prepare a brief note on the objectives of metrology and its relationship with standardization for the next lesson on calibration.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Importance of Calibration, classification of measurements, general care of instruments, objective of metrology, Standardization and various Standardizing organizations.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min